CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

November - 2012

CL 309 Construction Planning and Management

Date: 08 .11 .2012, Thursday **Time: 10:00 a.m. To 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Enlist the resources needed by construction industries. **02**

(b) A construction equipment was purchased at Rs. 1,80,000 /-. Assuming its salvage value at the end of 6 yr. to 10 % of purchase cost. Determine amount of depreciation for each year by Straight line method **05**

Q - 2 (a) State various objectives satisfied by construction management **02**

(b) The activity breakdown of certain project is a under. Prepare bar chart for the project. **06**

Activity No.	1	2	3	4	5	6	7
Duration (weeks)	1	2	4	3	1	2	4

Activity 2 & 3 can be performed concurrently and they both must follow activity 1

Activity 2 must precede activity 4

Activity 5 cannot begin until both activity 2 & 3 are complete

Activity 6 can be started only after activities 4 & 5 are complete

Activity 7 is the last activity which can be started only after completion of activity5

(c) What do you understand by work breakdown structure? What is its importance in network planning? **06**

OR

Q - 2 (a) Write short note on Engineer and construction project economy interaction **02**

(b) For the network shown in figure 1. Determine: Critical Path, Earliest and latest event time, Earliest and latest start and finish times, Free float and independent float for each activity. **06**

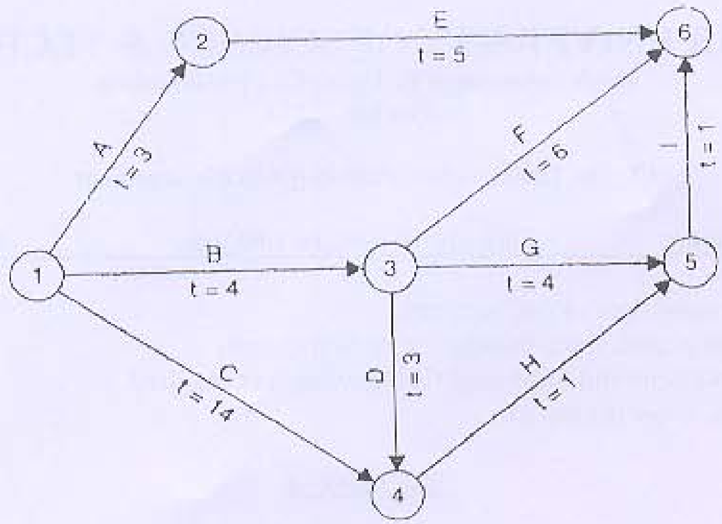

Figure 1

(c) Describe various steps for planning, scheduling and controlling a project. **06**

Q - 3 (a) What do you understand by the 'latest allowable occurrence time and earliest expected time? How do you determine it? **03**

(b) Define Head event, Tail event, Dual role event, Successor event, Predecessor event. **04**

(c) Plot the time-cost diagram for the project below : **07**

Activity	Duration (days)	Total cost	Activity	Duration (days)	Total cost
(1,2)	4	8000	(1,6)	2	5000
(2,3)	2	6000	(6,5)	5	10000
(2,4)	6	9000	(6,7)	7	7000
(3,5)	3	7500	(5,8)	1	4000
(4,5)	9	4500	(7,8)	10	1500

OR

Q - 3 (a) Explain various network rules. **07**

(b) Define the following : direct cost, indirect cost, outage loss, normal project time, normal cost, crash time, crash cost. **07**

SECTION – II

Q - 4 (a) State the activities and duties to be performed by government labour welfare officer. **02**

(b) In detail explain the types of organization. **05**

Q - 5 (a) Explain the Importance of management in Construction Industry. **06**

(b) Explain various phases of construction management. **04**

(c)	Which are the temporary services generally required on construction site?	04

OR

Q – 5 (a)	What is Project? State various aspects considered for Project formulation	06
(b)	Explain various causes of failures of construction projects.	04
(c)	How job layout influence the progress of project.	04
Q – 6 (a)	State three goal of project.	02
(b)	Write short note on Power Shovel – Consider type, size, selection and optimum depth of cut.	07
(c)	What is standard equipment? What are its advantages?	05

OR

Q – 6 (a)	Explain Objectives of management.	02
(b)	Describe scraper operation. How contractor can assure more production from scraper?	07
(c)	Explain the elements of owning cost of construction equipment.	05

****